



Hanzo Cloud is the sovereign AI operating layer for the new American industrial base — factories, robots, vehicles, sensors, drones, ships, and defense systems. We combine open-source AI infrastructure, spatial 3D, robotics control, post-quantum cryptography, FHE, and low-latency on-chain coordination so industrial and defense operators can **build, automate, simulate, secure, and command** real-world systems — from the factory floor to the contested edge — without depending on closed, foreign, or hyperscaler-controlled AI.

Why Hanzo

- **Sovereign industrial cloud.** An open-source, self-hostable AI cloud — agents, models, tools, memory, IAM/KMS, audit, sandboxes, deployment, secure workflows — running in cloud, factory, edge, or air-gapped. Not just chatbot infrastructure.
- **Spatial 3D + robotics.** Sits behind robotic cells, digital twins, spatial maps, autonomous vehicles, drones, inspection, and simulation-to-field workflows — exactly the Michigan industrial base of vehicles, autonomy, robotics, and ground systems.
- **Low-latency on-chain coordination.** Verifiable coordination *for machines*: identity, provenance, command logs, audit trails, access control, encrypted state, settlement, and trusted multi-party workflows between humans, robots, contractors, factories, and battlefield nodes.
- **Post-quantum + FHE-native security.** PQC secures comms and identity against future cryptographic risk; FHE lets agents, rules, and workflows compute *over ciphertext* — coalition analytics, cross-domain decisions, private supply-chain scoring, encrypted sensor fusion, private robotics telemetry.
- **Agentic engineering + cost collapse.** The same stack modernizes a contractor's software in place — fewer services, lower cloud cost, sovereign and auditable — as proven below.

Built for physical industry

- **Factory automation** — agentic workflows for production, QA, maintenance, scheduling, inventory, and process optimization.
- **Robotics & autonomy** — spatial 3D, simulation, perception pipelines, low-latency control, fleet coordination, and edge inference.
- **Defense systems** — secure command & telemetry, post-quantum identity, FHE-protected workflows, DDIL operation, and air-gapped deployment.
- **Industrial software modernization** — collapse legacy cloud, services, CI/CD, observability, auth, data, and security into an owned OSS stack.
- **On-chain machine coordination** — tamper-evident logs, provenance, access control, encrypted state, tasking, settlement, and inter-org trust.

The proof Founding CEO/CTO **Zach Kelling** was the **#1 Anthropic user by token count worldwide in 2025** (claudecount.com). That throughput built — with a small team on the owned OSS stack — a **FINRA/SEC-approved, SOC 2 securities exchange** on post-quantum crypto, a native-GPU blockchain, and FHE. Engaged for a narrow remediation scope, the team uncovered the real estate — **200+ microservices** on Google Cloud — and **consolidated it into 3 binaries**, deleting **3.5M lines of dead code** (3.07M→1.06M active) and closing **93 security findings** (22 critical). Cloud spend fell ~90% (~\$35K→~\$3.5K/mo, ~\$380K/yr saved). Backed by production post-quantum (FIPS 203/204/205), 50 machine-checked proofs, and the ZAP transport at **11.5M tx/s** (91% less memory, 96% less energy than JSON-RPC).

By the numbers: **11.5M tx/s** transport · **2.2 ns** parse · **91%** less memory · **96%** less energy · **30×** throughput vs JSON-RPC · **<2 min** rolling deploys · one **48 MB** binary replaces **17+** services (~27× smaller) · **~90%** cloud-cost cut (~\$380K/yr) · **200+→3** microservices · **3.5M** dead lines deleted · **93** security findings closed · **50** machine-checked proofs · **FIPS 203/204/205** in production · offline on embedded ARM.

For combat systems, Hanzo is the secure coordination layer: spatial awareness, low-latency agent control, post-quantum identity, encrypted telemetry, FHE-protected decision rules, and auditable on-chain command history — deployable at the edge when cloud links are degraded or denied.

The papers free to read & share at papers.hanzo.ai

Start here: **Sovereign vs. Closed** | **Agentic Engineering at Regulated Scale** | **Cloud Economics** | **Confidential Computing by Default (FHE)**

Depth: **ZAP** | **Full-Spectrum Cyber** | **Formal Verification** | **Autonomous Agentic AI** | **Edge AI/ML** | **FHE Cross-Domain** | **Assured Networking** | **Hanzo for Security** | **Hanzo for Compliance**



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